

Sai Mahesh Mudavat

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PROFESSIONAL SUMMARY

I am Ph.D. Candidate and Process-Driven Researcher specializing in Trustworthy Lightweight AI, Computer Vision, and Edge Computing for Smart Agriculture. Proven expertise in designing lightweight, resource-constrained models for on-device and edge AI which are intrinsically explainable, and Byzantine-robust neural architectures that bridge the gap between high-capacity deep learning and resource-constrained field robotics. Leverages a foundational industry background in scalable cloud orchestration and MLOps to ensure academic models are robust, deployable, and privacy-preserving. Moving forward, I am deeply interested in addressing challenges related to building hyper-localized AI models for agriculture, intercropping systems, and diverse agro-practices across different regions and environments.

EDUCATION

Ph.D. in Computer Science

University of North Texas

Aug 2024 – Present

Denton, TX

Master of Science, Computer Science

University of North Texas

Aug 2021 – May 2023

Denton, TX

Bachelor of Technology, Computer Science

Keshav Memorial Institute of Technology

Aug 2012 – May 2016

Hyderabad, India

RESEARCH INTERESTS

• Trustworthy Lightweight AI • Edge-Ready Computer Vision • Byzantine-Robust Federated Learning • Vision Foundation Models • Geometric Deep Learning • Explainable AI (XAI) • Linear-Cost State-Space Models • Mixed-Precision Quantization (INT4/INT8) • Group-Equivariant Convolutions • Physics-Informed Neural Networks (PINNs) • Generative AI • Retrieval-Augmented Generation (RAG) • Agentic AI Workflows • Natural Language Processing (NLP) • Cross-Crop Knowledge Graphs • Precision Agriculture

RESEARCH & INDUSTRY EXPERIENCE

Computer Vision Researcher & PhD student

University of North Texas (SESL)

Aug 2024 – Present

Denton, TX

- Designed a cross-architecture knowledge distillation pipeline (LiteViT) compressing a 300M-parameter Vision Transformer to a 3.8 MB MobileViT-XXS, achieving 99.3% accuracy at 12.26 ms inference latency on edge hardware.
- Architected a Byzantine-robust federated learning framework (DPRS) utilizing INT4/INT8 mixed-precision quantization, reducing bandwidth by 60% and sustaining 98.04% accuracy with automated failover via a Temporal Checkpoint Manifold.
- Developed a 28K-parameter dispersion-gated feedforward network (TinyVarNet) featuring parameter-free dynamic routing and intrinsic multi-scale spatial explainability, achieving 82.03% accuracy on highly cluttered real-field benchmarks.
- Engineered a sub-60K parameter C_8 group-equivariant convolutional network (E2-WeedNet) with a parameter-free Group-CAM operator to mitigate background shortcut learning and improve orientation-robust weed recognition.

- Built a 0.48M-parameter bidirectional Mamba state-space model (SAP-MambaNet) with learnable edge enhancement, successfully achieving 72.6% minority-class weed recall under severe field class imbalances ($< 0.1\%$ frame coverage).
- Formulated a few-shot localization framework (MAG-Net) pairing a frozen 22.1M-parameter DINOv2 foundation model with a 344K-parameter parametric Gabor stem, reaching 81.02% mIoU for cross-class weed generalization.

Software Engineer

Sep 2023 – Jul 2024

Adroit Affine

Dallas, TX

- Engineered and optimized backend Python and Java microservices, increasing overall system performance and throughput by 40%.
- Migrated monolithic applications to distributed containerized architectures utilizing Docker and Kubernetes, boosting scalability and cutting system downtime by 30%.
- Deployed scalable MLOps and CI/CD pipelines using Git, GitHub Actions, Jenkins, and ArgoCD, reducing model deployment time by 40%.
- Designed and integrated enterprise-grade Generative AI features using Retrieval-Augmented Generation (RAG), LLM orchestration, and Vector Databases for intelligent search applications.
- Built and deployed predictive machine learning models for Customer Operations via REST APIs, achieving a 15% improvement in data-driven decision-making efficiency.

TEACHING EXPERIENCE

Teaching Assistant - Fundamentals of AI

Aug 2026 – Dec 2026

University of North Texas

Denton, TX

- Guiding undergraduate students through core artificial intelligence concepts, state-space search algorithms, and intelligent agent design.

Teaching Assistant - Data Mining

Jan 2025 – May 2026

University of North Texas

Denton, TX

- Served as the primary TA for both graduate and undergraduate sections across three consecutive semesters.
- Designed and structured open-ended assignments, architected course materials, and created automated grading pipelines for comprehensive student evaluation.

Teaching Assistant - Computer Science II (C++)

Aug 2024 – Dec 2024

University of North Texas

Denton, TX

- Instructed laboratory sections focused on object-oriented programming, data structures, and dynamic memory management in C++.

Teaching Assistant - Machine Learning

Aug 2022 – May 2023

University of North Texas

Denton, TX

- Delivered guest lectures on deep learning fundamentals and core machine learning algorithms.
- Designed complex machine learning assignments and autonomously guided students through the complete lifecycle of their final predictive modeling projects using TensorFlow and PyTorch.

PUBLICATIONS

Published

- [1] **S. M. Mudavat**, A. Mitra, S. P. Mohanty, and E. Kougianos, “LiteViT: Leveraging the Power of Transformers for Edge AI in Crop Disease Classification,” in *Proceedings of the IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, 2025.

- [2] **S. M. Mudavat**, A. Mitra, S. P. Mohanty, and E. Kougianos, “FLitViT: Vision Transformers-Based Federated Learning Approach for Plant Disease Classification,” in *Proceedings of the 23rd OITS International Conference on Information Technology (OCIT)*, 2025.

Accepted

- [3] **S. M. Mudavat**, A. Mitra, S. P. Mohanty, and E. Kougianos, “DPRS: Secure and Lightweight Federated Vision Transformers for Agricultural Edge AI.”
- [4] **S. M. Mudavat**, A. Mitra, and S. P. Mohanty, “TinyVarNet: A Dispersion Gated Dual Kernel Network for Plant-Disease Classification.”
- [5] **S. M. Mudavat**, A. Mitra, and S. P. Mohanty, “E2-WeedNet: A Compact Equivariant Network for Shortcut-Aware Crop and Weed Recognition.”

ACADEMIC PROJECTS

- **Gesture Recognition System:** Deployed a scalable ML model on AWS EC2 with Docker containerization, achieving 95% real-time accuracy.
- **Video Q&A Platform:** Architected an LSTM-based encoder-decoder model coupled with NLP techniques to process video frames and natural language queries, enabling dynamic video-based question answering.
- **Telecom Churn Prediction & Friend Recommendation:** Implemented advanced machine learning pipelines and collaborative filtering techniques to predict customer churn with over 90% accuracy and boost recommendation relevance by 20%.

AWARDS & HONORS

- **Tuition Benefit Scholarship**, Toulouse Graduate School, University of North Texas, Aug 2024 – Present.

PROFESSIONAL SERVICE & ORGANIZATIONS

Professional Memberships:

- IEEE Member
- ACM Student Member

Peer Reviewer for Journals & Conferences:

- IFIP International Internet of Things (IoT) Conference
- IEEE Computer Society Annual Symposium on VLSI (ISVLSI)
- SN Computer Science (SNCS)
- VLSI Integration
- Earth Precision Agriculture (EPA)
- Applied Clinical Informatics
- ACM SIGCSE

CERTIFICATIONS

- Certificate of Excellence, “Certified Peer Reviewer Course,” Researcher Academy (2024)
- AWS Solutions Architect Associate (2023)
- Google Cloud Associate (2023)
- Terraform Associate (2023)

TECHNICAL SKILLS

- **Computer Vision & AI Models:** Semantic Segmentation, Object Detection, Image Classification, CNNs, State-Space Models (Mamba), Vision Transformers, Transfer Learning, Meta-Learning, Few-Shot Learning, Semi-Supervised Learning, Reinforcement Learning (RLHF).
- **Edge AI & Optimization:** Edge Deployment, On-Device Inference, TensorFlow Lite, TensorFlow.js, TinyML, Model Optimization, Pruning, Post-Training Quantization (INT4/INT8), ONNX, TensorRT, Embedded Systems.
- **Generative AI, NLP & Graphs:** Large Language Models (LLMs), RAG, Agentic Workflows, LangChain, LlamaIndex, Prompt Engineering, Named Entity Recognition, spaCy, Graph Neural Networks (GNNs), Knowledge Graphs, NetworkX.
- **Data Science & Algorithms:** PyTorch, TensorFlow, Keras, OpenCV, scikit-learn, Anomaly Detection, Clustering, Vector Databases (Pinecone, FAISS), PySpark, Pandas, NumPy.
- **DevOps, Cloud & MLOps:** Docker, Kubernetes (K8s), AWS, Azure, GCP, Git, GitHub Actions, Jenkins, ArgoCD, MLFlow Terraform, MLFLOW, CI/CD pipelines.
- **Languages & Architecture:** Python, C, C++, Java, JavaScript, TypeScript, Shell Scripting, SQL, FastAPI, REST APIs, Microservices, Spring Boot, Django, Flask, ROS 2, Linux.

REFERENCES

Name: Dr. Saraju P. Mohanty

Title: Professor

Address: Department of Computer Science and Engineering, University of North Texas, Denton, TX 76203.

Email: smohanty@ieee.org | **Phone:** (940) 565-3276 (Office)

Relationship: Major Advisor

Name: Dr. Alakananda Mitra

Title: Research Mentor

Address: Nebraska Water Center, University of Nebraska-Lincoln, Lincoln, NE 68583.

Email: amitra6@unl.edu

Relationship: Co-Major Advisor

Name: Dr. Venkata Prasanth Yanambaka

Title: Assistant Professor

Address: Department of Computer Science, Texas Woman's University, Denton, TX 76201.

Email: Pvyanambaka@gmail.com

Relationship: Mentor